Unsupervised Class-Guided Synthetic-to-Real Domain Adaptation for Plot-Level Forest UAV LiDAR Semantic Segmentation

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***Abstract*—Semantic segmentation of forest point clouds is pivotal for automated forest inventory but is severely constrained by the scarcity of high-quality real-world annotations. Although synthetic forest point clouds provide a scalable alternative for training, models trained on synthetic data often suffer from pronounced performance degradation when transferred to real forest scenes, due to domain shifts coupled with extreme structural heterogeneity and long-tailed semantic distributions. To address these challenges, we propose ForestPlotUDA, an unsupervised domain adaptation (UDA) framework specifically designed for plot-level forest UAV light detection and ranging (LiDAR) semantic segmentation. Unlike generic adaptation methods that rely on global feature alignment or dense pseudo-labeling, ForestPlotUDA explicitly targets the severe class imbalance and plot-wise structural variability inherent in forest environments, where ecologically critical woody components are sparsely distributed and easily overwhelmed by dominant foliage points. The proposed framework integrates decoupled feature normalization and class-guided self-training, enabling stable cross-domain adaptation under highly imbalanced and sparse supervision. Experimental results on the FOR-instance benchmark demonstrate that ForestPlotUDA improves the mean intersection-over-union (mIoU) from 54.71% to 61.50%. Notably, using only five pseudo-labeled points per class, the intersection-over-union (IoU) of the challenging wood class increases from 22.82% to 40.90%, highlighting the effectiveness of the proposed approach for annotation-free forest UAV LiDAR analysis. Additional experiments on SegmentedForests further demonstrate that ForestPlotUDA consistently improves over direct synthetic-to-real (Syn2real) transfer on ground-based terrestrial laser scanning (TLS)/mobile laser scanning (MLS) forest point clouds. These results indicate that explicitly accounting for structural heterogeneity and class imbalance is critical for Syn2Real adaptation in forest point clouds, paving the way for fully automated and low-cost forest inventory systems across diverse forest ecosystems.**

***Index Terms*—Forest point cloud, point cloud, segmentation, self-training, unsupervised domain adaptation (UDA).**

I. Introduction

FOREST ecosystems play a fundamental role in the global carbon cycle, biodiversity conservation, and climate regulation, and their accurate structural characterization has long been a central objective in forest ecology and remote sensing research [1]. Detailed knowledge of vertical and horizontal forest structure is essential for understanding forest organization and function and for characterizing ecosystem heterogeneity across scales [2].

Light detection and ranging (LiDAR) technologies have substantially advanced the observation and quantification of forest structure by providing accurate 3-D measurements of vegetation elements [3], [4], [5]. For example, terrestrial laser scanning (TLS) systems are capable of capturing fine-scale structural details from fixed ground positions, offering high point density and precise characterization of individual trees and understory vegetation [6]. Mobile laser scanning (MLS) systems extend this capability by enabling data acquisition along forest trails or roads, allowing efficient coverage of larger areas while maintaining sufficient spatial resolution [7]. Uncrewed aerial vehicle laser scanning (UAV-LS) has emerged as a complementary approach, providing rapid and cost-effective collection of plot-level forest point clouds with extensive spatial coverage and the ability to resolve canopy structure and vertical stratification [8], [9].

Despite the rich structural information provided by LiDAR observations, forest point clouds are inherently unstructured and consist of heterogeneous measurements from multiple vegetation components. Semantic segmentation, therefore, plays a pivotal role in transforming raw LiDAR point clouds into interpretable forest structural representations by assigning meaningful semantic labels to individual points [10], [11]. In this context, the reliable separation of ground, understory, foliage, and woody components constitutes a fundamental prerequisite for quantitative forest structural analysis, underpinning a wide range of downstream applications, such as structural characterization [12], biomass estimation [13], and the retrieval of ecological parameters [14].

In recent years, semantic segmentation of 3-D point clouds has achieved remarkable progress, largely driven by advances in deep learning architectures [15] and large-scale benchmark datasets [16]. A variety of convolution- and Transformer-based models have demonstrated strong performance in urban and indoor scenes by learning hierarchical geometric and contextual representations from densely annotated data [15], [17], [18], [19], [20]. These advances have also motivated increasing interest in extending point cloud semantic segmentation techniques to forest environments [21], [22], [23], [24]. However, compared with structured man-made scenes, forest

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point clouds exhibit substantially higher structural complexity, pronounced class imbalance, and strong variability across forest types, acquisition platforms, and sensing conditions. More critically, the development and evaluation of robust forest point cloud semantic segmentation models are severely constrained by the limited availability of high-quality, densely annotated forest point cloud datasets, which are costly and time-consuming to acquire in real-world conditions [23], [24], [25].

To address the persistent shortage of high-quality annotated forest point clouds, several complementary strategies have been explored in recent studies. These include the direct acquisition and manual or semiautomated annotation of real forest point clouds [23], [24], [25], annotation-efficient approaches based on weakly supervised, semi-supervised, or interactive learning [26], [27], as well as emerging unsupervised segmentation methods that avoid the need for labeled data altogether [28], [29]. Among these directions, the generation of synthetic forest point clouds through procedural modeling or simulation has attracted increasing attention, as it enables the construction of large scale, fully annotated datasets that can be used to pretrain models or supplement scarce real data [30], [31], [32], [33], [34], [35]. In particular, synthetic-to-real (syn2real) approaches have demonstrated notable potential by allowing models trained on synthetic data to transfer knowledge to real forest environments [36].

However, despite the advantages of Syn2Real approaches, models trained on synthetic point clouds often suffer from pronounced domain gaps when deployed in real-world environments [37], [38]. These gaps are particularly severe in forest environments due to the inherent structural complexity and variability of vegetation, as well as differences across forest types and acquisition conditions [39], [40]. Unsupervised domain adaptation (UDA) and related transfer learning techniques have been shown to partially mitigate these discrepancies, but existing efforts are largely limited to tree-level or part-level segmentation with a restricted set of semantic components [41]. In contrast, plot-level forest point clouds encompass multiple interacting semantic classes, including ground, understory vegetation, foliage, and woody elements, which substantially amplifies domain gaps and renders knowledge transfer more challenging. Consequently, a systematic investigation of UDA strategies tailored to plot-level forest point cloud semantic segmentation remains largely unexplored, motivating the present study.

To bridge this gap, this article proposes a unified UDA framework, ForestPlotUDA, specifically tailored for the semantic segmentation of plot-level forest point clouds. By leveraging high-fidelity synthetic data as a source domain, our approach aims to transfer semantic knowledge to unlabeled real-world forest plots while overcoming the specific challenges of complex vegetation structures. Unlike general-purpose UDA methods, ForestPlotUDA explicitly targets the dual hurdles of extreme class imbalance where woody components are vastly outnumbered by ground and foliage, and the distinct distributional shifts arising from heterogeneous forest types. Through a synergy of decoupled normalization and class-balanced feature alignment, the proposed framework achieves robust segmentation performance without requiring manual annotations on target real-world data. The main contributions of this study can be summarized as follows.

1) We propose ForestPlotUDA, the first UDA framework specifically tailored for plot-level forest point clouds, effectively bridging the Syn2Real gap without manual annotations.
2) We design three domain-specific components, including plot-decoupled normalization, class-guided point selection (CGPS), and category-aware memory bank (CAMB), to synergistically align feature distributions and tackle extreme class imbalance.
3) Extensive experiments demonstrate that our method significantly outperforms state-of-the-art UDA baselines and Syn2Real pretraining, showing robust transferability across heterogeneous forest types, particularly on minority classes.

II. Related Work

A. UDA for Point Clouds

UDA for point cloud semantic segmentation arises from distribution shifts induced by heterogeneous acquisition conditions, sensor configurations, and scene structures. Motivated by this objective, prior studies have explored a wide range of strategies, including distribution alignment, knowledge transfer, and explicit modeling of scene- and sensor-specific characteristics [42], [43], [44], [45].

One line of research seeks to reduce domain discrepancies by enforcing distributional consistency between the source and target domains, typically through adversarial objectives that encourage domain-invariant intermediate representations [39], [43], [44], [46], [47], [48], [49]. Some methods perform alignment in the input space, where point sampling patterns are reshaped, or sensor noise is simulated to reduce measurement-level discrepancies [46]. Others conduct alignment in the feature space, constraining semantic representations to be domain-invariant through discriminative or adversarial losses [47]. Although global alignment strategies are effective in alleviating sensor-induced variations, they often neglect the heterogeneity of semantic classes in spatial structure and scale, which may diminish class-specific geometric information and degrade discriminability. To address this limitation, recent studies have proposed category-aware or local structural alignment mechanisms that operate at finer granularity, thereby reducing semantic interference during cross-domain adaptation [44], [48], [49].

Another prominent research direction leverages latent supervisory signals in the unlabeled target domain through self-training [50], [51], [52], [53], [54]. In these approaches, pseudo-labels generated by a source-trained model are iteratively refined, enabling the segmentation network to progressively adapt to the target-domain data. To mitigate noisy pseudo-labels and structural inconsistencies, prior works introduce regularization strategies such as temporal or multiview consistency and cross-domain data composition [50], [51], [52], [53]. Despite demonstrating strong empirical performance on standard benchmarks, self-training approaches

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remain sensitive to the initialization of the source model and the accumulation of pseudo-label errors, particularly in scenarios characterized by substantial variations in point density and geometric complexity [54].

B. Transfer Learning for Forest Point Clouds

In forest point cloud analysis, acquiring large-scale and fine-grained annotations is costly and time-consuming, which has motivated the adoption of transfer learning to reduce reliance on labeled target data. Existing studies can be broadly grouped into three complementary research directions.

The first direction focuses on representation pretraining using large-scale data. Self-supervised learning on unlabeled forest point clouds and pretraining with synthetic data generated by forest simulators have been explored to learn transferable geometric representations that improve robustness under limited supervision [55], [56], [57]. However, these approaches primarily target representation generality and do not explicitly address cross-forest domain shifts. The second direction investigates transfer learning through fine-tuning with limited labeled data. By adapting pretrained models to target datasets with a small number of annotations, these methods achieve notable performance gains for tasks such as individual tree segmentation and forest structural analysis, while still relying on some degree of target supervision [21], [58]. Their effectiveness may degrade when annotation budgets are extremely constrained or when forest types differ substantially. A third line of research addresses fully UDA, where the target domain contains no labeled data and exhibits distributional discrepancies relative to the source domain. Existing studies attempt to learn domain-invariant representations through feature alignment or pseudo-label-based knowledge transfer, enabling adaptation without supervision [41], [59]. Nevertheless, these methods have largely been developed for object-level or controlled settings and have not been systematically evaluated for plot-level forest semantic segmentation.

While the above studies demonstrate the effectiveness of transfer learning for forest point cloud analysis under different supervision regimes, they primarily focus on representation reuse or limited adaptation across datasets. However, these approaches have not been systematically investigated at the plot or stand level in forest environments, where substantial differences across forest types and extreme imbalance among semantic categories pose additional challenges. Motivated by this gap, we propose a UDA framework tailored to forest point cloud semantic segmentation that explicitly accounts for interforest structural variation and highly imbalanced class distributions.

III. Materials and Methods

A. Datasets

In this study, we evaluate ForestPlotUDA under two Syn2Real adaptation settings. The primary setting uses Boreal3D-ULS [36] as the labeled synthetic source domain and FOR-instance [25] as the unlabeled real UAV-LS target domain. To further evaluate transferability to ground-based forest point clouds, an additional setting uses Boreal3D-MLS [36] as the labeled synthetic source domain and SegmentedForests [60] as the unlabeled real ground-based TLS/MLS target domain. For all datasets, we follow the training and test splits defined in their original publications. In this section, detailed descriptions of the synthetic source data, real target data, and label harmonization protocols are provided.

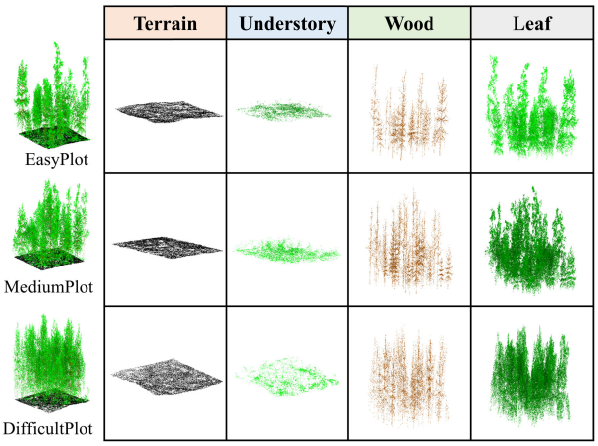


Fig. 1. Visualization of four semantic classes from Boreal3D forest plots with varying structural complexities.

1) Synthetic Forest Point Cloud: The Boreal3D dataset [36] is a large-scale synthetic forest point cloud dataset generated using procedural modeling and comprises 1000 boreal forest plots. Each plot contains a mixture of three common boreal tree species, namely spruce, pine, and birch, reflecting typical species composition in boreal forest stands. All points are annotated with point-wise semantic labels corresponding to *terrain*, *understory*, *wood*, and *leaf*, together with instance-level labels identifying individual trees. In addition, plot-level structural attributes, including tree height and diameter at breast height, are provided.

In the experiments, the ULS subset is used for the main UAV-LS adaptation setting, while the MLS subset is used for the cross-platform validation on ground-based forest point clouds. Following the original Boreal3D scene configurations, the synthetic forest plots exhibit varying levels of structural complexity, which are illustrated in Fig. 1 using representative examples.

2) Real Forest Point Cloud: Two real forest point cloud benchmarks are used as target domains in this study. FOR-instance [25] is used as the UAV-LS target benchmark. It is a publicly available dataset collected over five geographically distinct forest sites representing diverse forest types and structural conditions, including CULS, NIBIO, TUWIEN, RMIT, and SCION. The dataset provides high-density 3-D point clouds with point-wise semantic and instance-level annotations, enabling the characterization of forest structural components from an aerial perspective. In its original annotation scheme, FOR-instance defines seven semantic classes, including *unclassified*, *low-vegetation*, *terrain*, *out-points*, *stem*, and *live branches*, with *woody branches* annotated separately.

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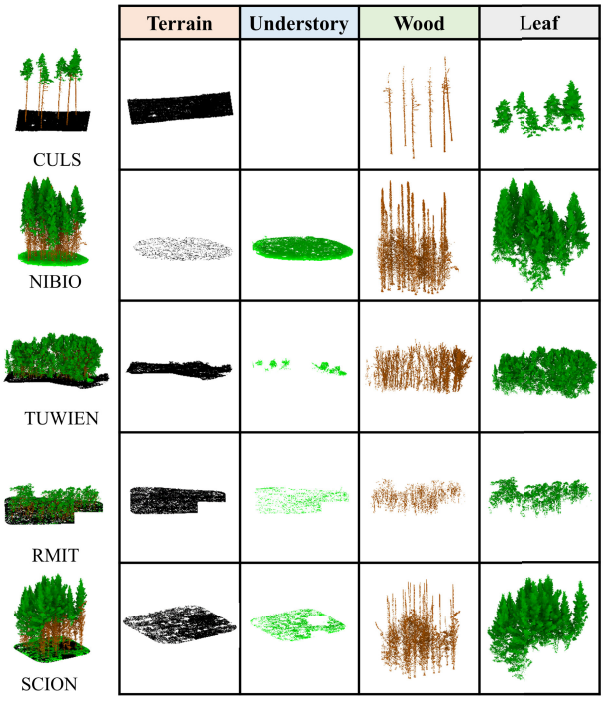


Fig. 2. Visualization of four semantic classes from five subsets of the FOR-instance forest plots.

To ensure consistency with widely used forest semantic segmentation settings, the original FOR-instance labels are harmonized into four ecologically and structurally meaningful categories: *terrain*, *understory*, *wood*, and *leaf*. Specifically, points labeled as *unclassified* and *out-points* are excluded from further analysis. The original *terrain* and *low-vegetation* classes are mapped to *terrain* and *understory*, respectively. The *stem* and *woody branches* classes are merged into a unified *wood* category, while *live branches* are reclassified as *leaf*. Fig. 2 presents representative visualizations of the relabeled semantic components for each forest site.

SegmentedForests [60] is used as a ground-based TLS/MLS target benchmark to further evaluate transferability beyond UAV-LS data. It is a recently released dataset of manually annotated ground-based forest point clouds designed for semantic segmentation. It contains more than 920 million labeled points from 14 forest plots acquired using TLS and MLS. The dataset covers diverse broadleaf, coniferous, and mixed forest stands from multiple bioclimatic regions and includes more than 1600 trees across 16 tree species.

The original SegmentedForests annotations contain multiple vegetation and nonvegetation categories. To maintain consistency with the four-class semantic setting used in this study, we remap the original labels into four target categories: *terrain*, *understory*, *wood*, and *leaf*. Specifically, *ground* + *ground vegetation* is mapped to *terrain*; *shrub* is mapped to *understory*; *stem*, *understory tree stem*, and *lower canopy tree stem* are mapped to *wood*; and *branches* + *leaves*, *understory tree branches*, and *lower canopy tree branches* are mapped to *leaf*. Other categories, including dead downwood, stumps, artificial objects, people, rocks, standing dead stems, ivy, and other unclassified elements, are excluded from evaluation.

It should be noted that, unlike FOR-instance, SegmentedForests does not explicitly separate woody branches from foliage in all tree components. Consequently, the *wood* class in SegmentedForests mainly corresponds to stem-related structures, while branch-and-leaf components are grouped into the *leaf* category.

B. ForestPlotUDA

1) Overall Syn2Real Self-Training Framework for Forest Point Cloud Segmentation: We propose a Syn2Real self-training framework for semantic segmentation of forest point clouds acquired by UAV LiDAR systems. The framework adopts a teacher–student paradigm, where both networks share the same backbone architecture based on Point Transformer v3 (PTv3) [20]. As illustrated in Fig. 3, the framework takes two types of data as input during training: fully annotated synthetic forest point clouds and unlabeled real forest point clouds. The objective of the framework is to transfer the knowledge learned from synthetic forest scenes to real-world UAV point clouds through self-training.

The backbone network follows the standard encoder–decoder architecture of PTv3 and is responsible for point-wise feature extraction from forest point clouds. For each input point cloud, several preprocessing steps are applied, including center shifting, random rotation along the x-, y-, and z-axes, and random scaling. The processed point cloud is then voxelized. Instead of averaging point coordinates within each voxel, a single point is randomly sampled to represent the voxel. To control computational cost, region cropping is applied, and cylindrical sampling is used to limit the spatial extent of each input region. The preprocessed point clouds are directly fed into the backbone to obtain per-point feature representations. Within the backbone, point features are first initialized by computing three different point orderings following the standard PTv3 setting [20], after which the first transformer block is applied. The encoder and decoder share the same building blocks but are configured asymmetrically. The encoder consists of four stages with block numbers set to 2, 2, 6, and 4, respectively, while the decoder uses a lighter configuration with one block per stage. Grid pooling is applied to model local spatial relationships among neighboring points, followed by a shuffle operation on point orderings. The extracted features are finally passed to the segmentation head to produce point-wise semantic prediction probabilities.

During training, the backbone is first pretrained solely on synthetic forest point clouds with ground-truth annotations. After pretraining, the learned weights are used to initialize both the teacher and student networks. Plot-decoupled feature normalization modules (see Section III-B2) are incorporated into the shared backbone. The teacher is then applied to each real point cloud to generate per-point class probabilities and associated feature embeddings. A class-guided point selection (CGPS) module (see Section III-B3) filters high-confidence pseudo-labeled points based on their predicted class probabilities, producing a set of selected pseudo-labels along with their associated features. The selected point features are subsequently stored in a category-aware memory bank

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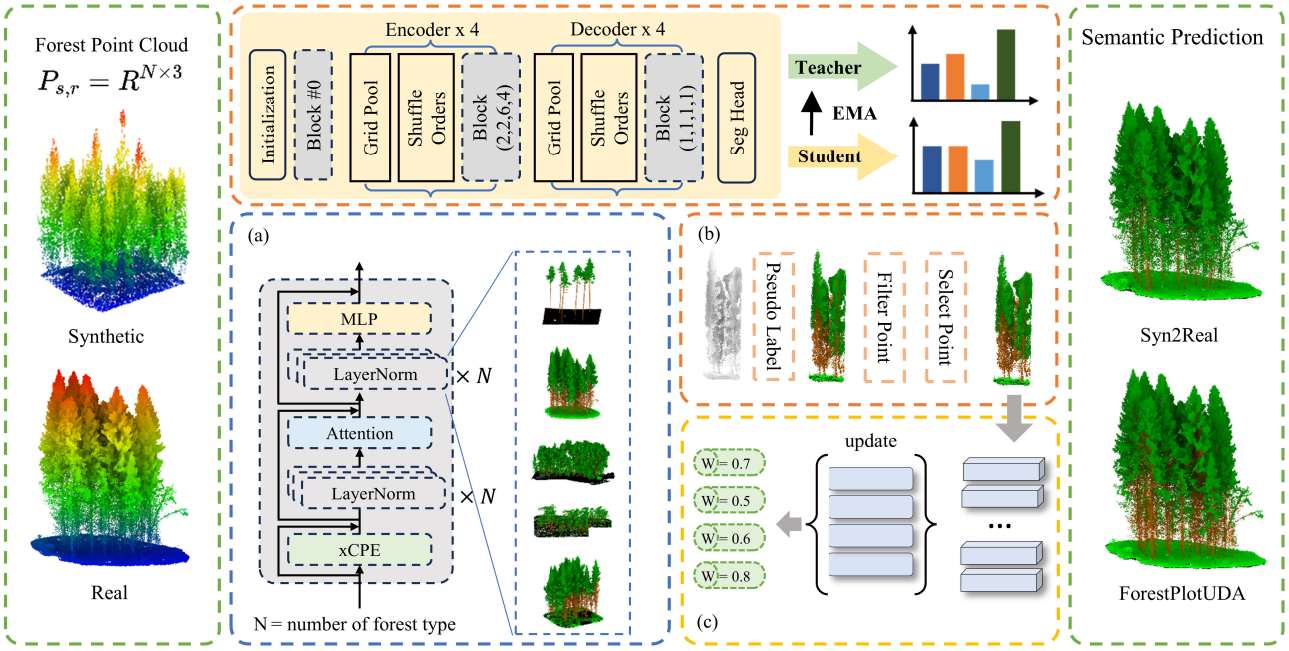


Fig. 3. Overall architecture diagram of ForestPlotUDA. The framework follows a teacher–student paradigm and consists of three main components. (a) Plot-decoupled normalization, (b) CGPS, and (c) CAMB.

(CAMB), which maintains class-specific feature representations and predicts confidence-aware weights for the selected pseudo-labels (see Section III-B4). The student network is optimized using the selected pseudo-labeled points with a weighted cross-entropy loss and a Lovász loss. During self-training, the teacher network is updated as an exponential moving average (EMA) of the student parameters.

2) Plot-Decoupled Feature Normalization for Heterogeneous Forest Structures: Forest plots differ markedly in canopy density, species composition, vertical stratification, and point sampling density under UAV LiDAR acquisition. These per-plot and per-type differences lead to substantial shifts in intermediate feature statistics, which in turn can bias pseudo-label generation and destabilize adaptation when a single, shared normalization is applied across heterogeneous plots. Normalization layers inside a deep network play a central role in controlling feature distributions and stabilizing optimization [61]. In the context of plot-level forest segmentation, conditioning normalization on plot or type identity can prevent destructive interference between feature statistics originating from structurally different plots (as illustrated in Fig. 2, the five forest types exhibit markedly different spatial point distributions), thereby producing more consistent semantic predictions within each plot.

To address this issue, we adopt a plot-decoupled normalization strategy inspired by prompt-driven normalization (PDNorm) [62]. Instead of conditioning normalization on learnable prompts, we instantiate the prompt as the forest plot type and assign each type an independent LayerNorm module. Given intermediate point-wise features $\mathbf{h} \in \mathbb{R}^{N\times d}$ extracted from a single forest plot of type t, normalization is performed as

$$\tilde{\mathbf{h}}_n = \frac{\mathbf{h}_n - \boldsymbol{\mu}_t}{\sqrt{\boldsymbol{\sigma}_t^2 + \epsilon}}, \quad n = 1, \ldots, N \tag{1}$$

where the mean and variance are computed over all points within the same plot

$$\boldsymbol{\mu}_t = \frac{1}{N}\sum_{n=1}^{N}\mathbf{h}_n, \quad \boldsymbol{\sigma}_t^2 = \frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{h}_n - \boldsymbol{\mu}_t\right)^2. \tag{2}$$

Each forest type maintains its own independent normalization statistics, while no additional learnable parameters are introduced beyond those already present in standard LayerNorm. As illustrated in Fig. 3(a), the proposed module is embedded into the backbone network and selectively applies the corresponding normalization operation according to the plot identity, while all other network parameters remain shared. During both training and inference, the plot identity determines which normalization module is applied. The plot identity is obtained from acquisition-level metadata or file organization and does not require point-wise semantic annotations. This design decouples feature normalization across heterogeneous forest plots while preserving internal consistency within each plot, thereby reducing cross-plot interference under UDA.

3) CGPS for Structurally Imbalanced Forest Scenes: Forest point clouds collected by UAV-LS are typically dominated by foliage points, while structurally important classes such as woody elements and understory are sparse and often occluded. Under such conditions, naively using all high-confidence pseudo-labels would result in supervision that is heavily biased toward dominant canopy points, providing limited learning signal for minority classes.

To address this issue while keeping the selection strategy simple and robust, we adopt a two-step point selection procedure. As shown in Fig. 3(b), first, a confidence threshold τ is applied to the teacher predictions to retain only reliable

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pseudo-labels

$$\mathcal{P} = \left\{ n \mid \max_c \mathbf{p}_{t,n}^{(c)} \geq \tau \right\}. \tag{3}$$

This step filters out low-confidence predictions that are more likely to be affected by occlusion, mixed returns, or domain shift between synthetic and real forest scenes.

Second, to explicitly control class imbalance during self-training, a fixed number K of points is randomly sampled for each semantic class from the high-confidence set. Let $\mathcal{P}_c$ denote the subset of points in $\mathcal{P}$ predicted as class c. The final pseudo-labeled supervision set is defined as

$$\mathcal{S} = \bigcup_{c=1}^{C} \text{RandomSample}_K(\mathcal{P}_c). \tag{4}$$

Random sampling within each class serves two purposes in forest scenes. First, it prevents the supervision from being dominated by spatially clustered canopy points, which often exhibit uniformly high confidence. Second, it encourages exposure to diverse local structures, including thin woody elements and fragmented understory vegetation, without introducing additional heuristic assumptions. If fewer than K points are available for a given class, all available high-confidence points are retained.

4) CAMB for Pseudo-Labeled Forest Points: As discussed in Section III-B3, our framework deliberately adopts an extremely sparse pseudo-labeling strategy by randomly sampling a fixed number of high-confidence points per class. It also introduces an additional source of training instability due to the stochastic nature of point selection, particularly in early adaptation stages. To further stabilize optimization under such sparse supervision, we introduce a prototype-guided soft weighting mechanism that adaptively modulates the contribution of individual pseudo-labeled points. As shown in Fig. 3(c), the memory bank maintains a class-specific feature prototype for each semantic category, which captures representative feature directions in the real-domain feature space.

Let $\mathbf{h}_n \in \mathbb{R}^d$ denote the L_2-normalized feature of point n predicted as class $\hat{y}_n$. For each class c, a prototype vector $\mathbf{m}_c \in \mathbb{R}^d$ is maintained and updated using an EMA

$$\mathbf{m}_c \leftarrow (1 - \mu)\, \mathbf{m}_c + \mu\, \text{Median}\left(\{\mathbf{h}_n \mid \hat{y}_n = c\}\right) \tag{5}$$

where μ is the momentum coefficient, and the median operation is applied element-wise to reduce sensitivity to noisy pseudo-labels.

Given a pseudo-labeled point n, its reliability weight is computed based on the cosine similarity between its feature and the corresponding class prototype

$$s_n = \sigma\left(\frac{\mathbf{h}_n^\top \mathbf{m}_{\hat{y}_n}}{T}\right) \tag{6}$$

where T is a temperature parameter controlling the sharpness of similarity responses and $\sigma(\cdot)$ denotes the sigmoid function.

5) Training Objective: The student network is optimized using a composite loss function consisting of a weighted cross-entropy loss and a Lovász–Softmax loss

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Lovász}}. \tag{7}$$

The cross-entropy term enforces point-wise semantic consistency under pseudo-label supervision and is modulated by the prototype-guided soft weights described in Section III-B4

$$\mathcal{L}_{\text{CE}} = \sum_{n \in \mathcal{S}} \tilde{s}_n\, \ell_{\text{CE}}\left(\mathbf{p}_{s,n}, \hat{y}_n\right). \tag{8}$$

The Lovász–Softmax loss directly optimizes a differentiable surrogate of the mean intersection-over-union (mIoU) across semantic classes

$$\mathcal{L}_{\text{Lovász}} = \mathcal{L}_{\text{Lovász}}\left(\mathbf{p}_s, \hat{y}\right) \tag{9}$$

where $\mathbf{p}_s$ denotes the student predictions after Softmax normalization. The Lovász loss is computed without sample-wise reweighting, allowing it to preserve its ranking-based formulation and emphasize global structural consistency.

C. Evaluation Metrics

The performance of the proposed framework for forest point cloud semantic segmentation was evaluated using three standard metrics addressing class imbalance and structural complexity in forests.

1) *Overall Accuracy (OA)*: Proportion of correctly classified points

$$\text{OA} = \frac{\sum_{c=1}^{C} \text{TP}_c}{N}. \tag{10}$$

2) *Mean Accuracy (mAcc)*: Average per-class accuracy

$$\text{mAcc} = \frac{1}{C} \sum_{c=1}^{C} \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}. \tag{11}$$

3) *mIoU*: Average class-wise intersection over union (IoU)

$$\text{IoU}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{mIoU} = \frac{1}{C} \sum_{c=1}^{C} \text{IoU}_c. \tag{12}$$

where C = number of classes (e.g., terrain, understory, wood, and leaf), N = total number of points, and for class c:

1) TP_c = correctly classified points;
2) FP_c = points misclassified as class c;
3) FN_c = ground-truth points of class c missed.

mIoU is our primary metric due to its sensitivity to class-wise segmentation quality in complex forest structures.

D. Implementation Details

Raw point clouds were voxelized with a voxel size of 0.2 m. Training samples were generated by randomly cropping cylindrical patches with a radius of 4.0 m. Data augmentation included random rotation around the vertical axis, uniform scaling, and coordinate jittering. The network was optimized using the AdamW optimizer with an initial learning rate of 1×10^{-4} and a weight decay of 0.05. Parameters associated with backbone blocks were assigned a reduced learning rate of 1×10^{-5}. A cyclic learning rate schedule with a warmup ratio of 0.05 and cosine annealing was used. The model was pretrained on synthetic data for 500 epochs, followed by UDA training for 100 epochs. During target-domain adaptation, the

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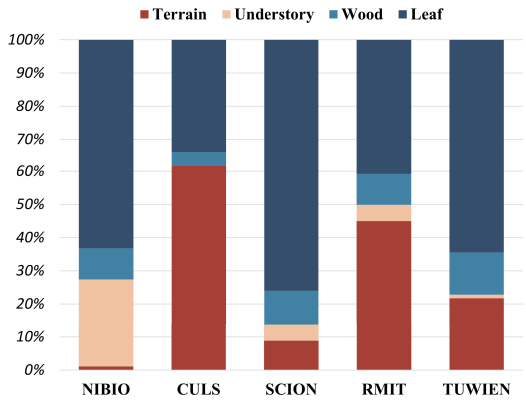


Fig. 4. Stacked distribution of semantic point components across five subsets of the FOR-instance forest plots, illustrating the relative dominance and imbalance of different semantic categories.

model was trained for a fixed number of epochs, and the checkpoint from the last adaptation epoch was used for final testing.

Experiments were conducted on a workstation with a 192-core Intel Xeon Platinum 8475B processor, 512 GB of DRAM, and four NVIDIA RTX 4090 GPUs (24 GB each) running Ubuntu 22.04. Our implementation is built upon and trained within the Pointcept framework (https://github.com/Pointcept/Pointcept) with PyTorch 2.6.0 and CUDA 11.8. The source code and experimental configurations are publicly available at https://github.com/ijliu/ForestPlotUDA

IV. Results

A. Forest Plot Characteristics and the Influence of Class Imbalance on Semantic Segmentation

Fig. 4 shows the relative proportions of the four semantic components across the forest plots in the FOR-instance dataset. Within individual plots, the distributions are uneven, with the leaf class occupying the largest proportion of points in most plots, while terrain and understory account for comparatively smaller shares. For example, in the SCION plot, the uneven dominance of the leaf class is particularly marked. The wood class, representing a principal structural component, remains relatively rare across all plot types, with a comparatively low share of points. The class composition varies across forest plots. The understory class is absent in the CULS plot, whereas it is present in NIBIO and SCION with different relative proportions. The contributions of wood and leaf also differ across plots, producing distinct class distributions for each forest plot.

To examine how distributional heterogeneity across forest plots affects model optimization and segmentation performance, Table I reports quantitative results under different training settings. The *single-plot (unified)* results are reported from the Boreal3D benchmark, where models are first pre-trained on synthetic data and subsequently fine-tuned on real forest point clouds with ground-truth annotations, and performance is reported separately for each forest type. These results serve as a reference upper bound for plot-level semantic segmentation performance under ideal supervision.

TABLE I

mIoU Under Different UDA Settings. Single Plot Denotes Plot-Specific Adaptation Using Pseudo-Labels From the Corresponding Plot

Schemes	Plot	mIoU (Unified)	mIoU (Plot-level)
Single Plot	CULS	**66.13**	61.10
	NIBIO	**68.44**	57.00
	RMIT	50.69	**57.84**
	SCION	52.70	**54.07**
	TUWIEN	**55.99**	53.53

In contrast, when pseudo-labeled samples from all plots are directly mixed for UDA, the training process becomes unstable and fails to converge, as indicated by the divergence of the training loss. This behavior suggests that naively aggregating forest plots with drastically different class distributions can severely disrupt optimization, particularly when minority classes are sparsely represented or entirely absent in certain plots. When UDA is conducted independently on individual plots (*single plot*), stable convergence is consistently achieved. Under this setting, models are trained using plot-specific pseudo-labels and evaluated on the corresponding plots. As shown in Table I, the resulting plot-level mIoU values are comparable to the fully supervised reference in several cases. For example, on RMIT and SCION, plot-level UDA even exceeds the reported unified mIoU, while on CULS and NIBIO the performance gap remains moderate.

B. Quantitative Comparison With State-of-the-Art Methods

To assess the effectiveness of our method in the context of plot-level forest point clouds, we selected representative UDA baselines covering the major adaptation strategies. ADVENT [43] employs adversarial feature alignment, serving as a representative of distribution-level alignment approaches. CosMix [53] and PolarMix [52] implement self-training with point cloud mixing. Specifically, Table II presents a quantitative comparison of different training strategies for plot-level forest point cloud segmentation on real UAV LiDAR point clouds. To obtain detailed class-wise metrics under the revised training protocol, we reproduced the original Real2Real and Syn2Real settings. Rows marked with * denote the reproduced results. The Real2Real setting, corresponding to fully supervised training and evaluation on real data, serves as an upper-bound reference and achieves the highest overall performance, with an mIoU of 81.48%. In contrast, the Syn2Real baseline, where the model is trained exclusively on synthetic data and directly evaluated on real-world scenes, exhibits a pronounced performance degradation, with mIoU dropping to 54.71%. This decline is particularly evident for the terrain and wood classes, highlighting the substantial domain discrepancy between synthetic and real forest point cloud distributions. ADVENT [43] achieves moderate mIoU but shows particularly poor performance on the wood class, with IoU close to zero. This suggests that global adversarial feature alignment alone is insufficient to handle the severe structural heterogeneity and

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TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT TRAINING STRATEGIES FOR PLOT-LEVEL PLANT STRUCTURE SEGMENTATION (UNIT: %). ROWS MARKED WITH * DENOTE REPRODUCED RESULTS

Method	OA	mAcc	mIoU	Terrain	Understory	Wood	Leaf
Real2Real	92.69	85.80	78.59	-	-	-	-
Real2Real*	93.34	90.15	81.48	80.23	89.47	63.52	92.72
Syn2Real	84.41	66.02	55.28	-	-	-	-
Syn2Real*	83.84	65.72	54.71	40.95	67.49	22.82	87.60
ADVENT [43]	-	-	30.00	30.3	10.9	0.6	78.5
CosMix [53]	14.56	35.26	8.94	6.36	15.68	12.84	0.17
PolarMix [52]	-	-	37.19	33.27	30.36	0.19	84.95
Weight	**85.72**	66.66	56.90	**45.94**	72.95	20.16	**88.56**
Our	85.58 (↑ 1.74)	**73.82** (↑ 8.10)	**61.50** (↑ 6.79)	44.26 (↑ 3.31)	**73.07** (↑ 5.58)	**40.90** (↑ 18.08)	87.78 (↑ 0.18)

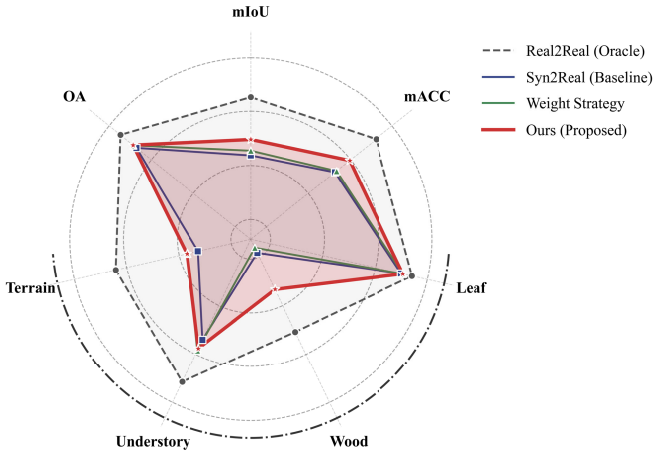


Fig. 5. Radar plot comparing class-wise IoU for representative training strategies on real forest point clouds.

sparsity of forest point clouds. While it can partially adapt terrain and leaf representations, the pronounced interclass imbalance and complex tree structures limit its effectiveness in plot-level forest segmentation. Several existing point cloud mixing and augmentation strategies were further evaluated under the same experimental protocol. Both CosMix [53] and PolarMix [52] fail to achieve stable convergence in our setting. CosMix yields extremely low accuracy across all semantic categories, while PolarMix demonstrates highly uneven class-wise performance, with near-zero accuracy for the wood component. These results indicate that data mixing strategies originally developed for urban or object-centric point clouds are not readily transferable to plot-level forest environments, where class imbalance and structural heterogeneity are substantially more severe.

To mitigate the impact of class imbalance during UDA, a class-weighted self-training baseline was also evaluated. While this strategy improves overall performance relative to the Syn2Real baseline, achieving an mIoU of 56.90%, its gains remain limited, particularly for the wood class. In comparison, the proposed method consistently improves segmentation accuracy across most semantic categories, reaching an mIoU of 61.50% and a notably higher mean class accuracy. Beyond OA metrics, Fig. 5 provides a radar-based visualization of class-wise IoU profiles for representative training strategies. The radar plot highlights a more balanced performance distribution achieved by the proposed method, especially for structurally underrepresented components such as wood. In contrast, both the Syn2Real baseline and existing augmentation-based methods exhibit highly skewed performance profiles, dominated by foliage segmentation while failing to adequately capture woody structures.

C. Qualitative Visualization on Real Forest Point Clouds

To qualitatively assess the effectiveness of the proposed Syn2Real adaptation framework, we visualize semantic segmentation results on six representative forest plots from the FOR-instance dataset. For each plot, side-by-side comparisons of ground-truth (GT) annotations, the direct Syn2Real baseline, and the proposed method (Ours) are presented. As shown in Fig. 6, each plot is arranged in four rows: a global overview of the entire plot, followed by three zoomed-in local regions selected to highlight fine-scale structural details and challenging segmentation scenarios.

Across all visualized plots, the most prominent failure mode of the Syn2Real baseline is observed in the *wood* class. In both global views and local regions, a substantial portion of woody components are incorrectly classified as *Leaf*, resulting in fragmented or blurred representations of tree stems and branches. While misclassifications are also present for the *terrain* and *understory* classes, such errors are generally less visually salient and less consistently observable across plots. Syn2Real further exhibits unstable behavior in understory segmentation for certain plots. For instance, in the second zoomed-in region of the TUWIEN plot, understory vegetation is largely misclassified as leaf or woody components, leading to an ambiguous vertical structure. In contrast, the proposed method produces noticeably more coherent segmentation results for woody components across all plots. Many woody structures that are misclassified by Syn2Real are correctly recovered, forming more continuous and spatially consistent representations of tree stems. Although differences with the ground truth remain at fine scales, the overall improvement is clearly visible in both global and local views.

Notably, in regions where woody components are spatially intertwined with dense foliage, Syn2Real exhibits a strong tendency to assign leaf labels to the majority of points. This

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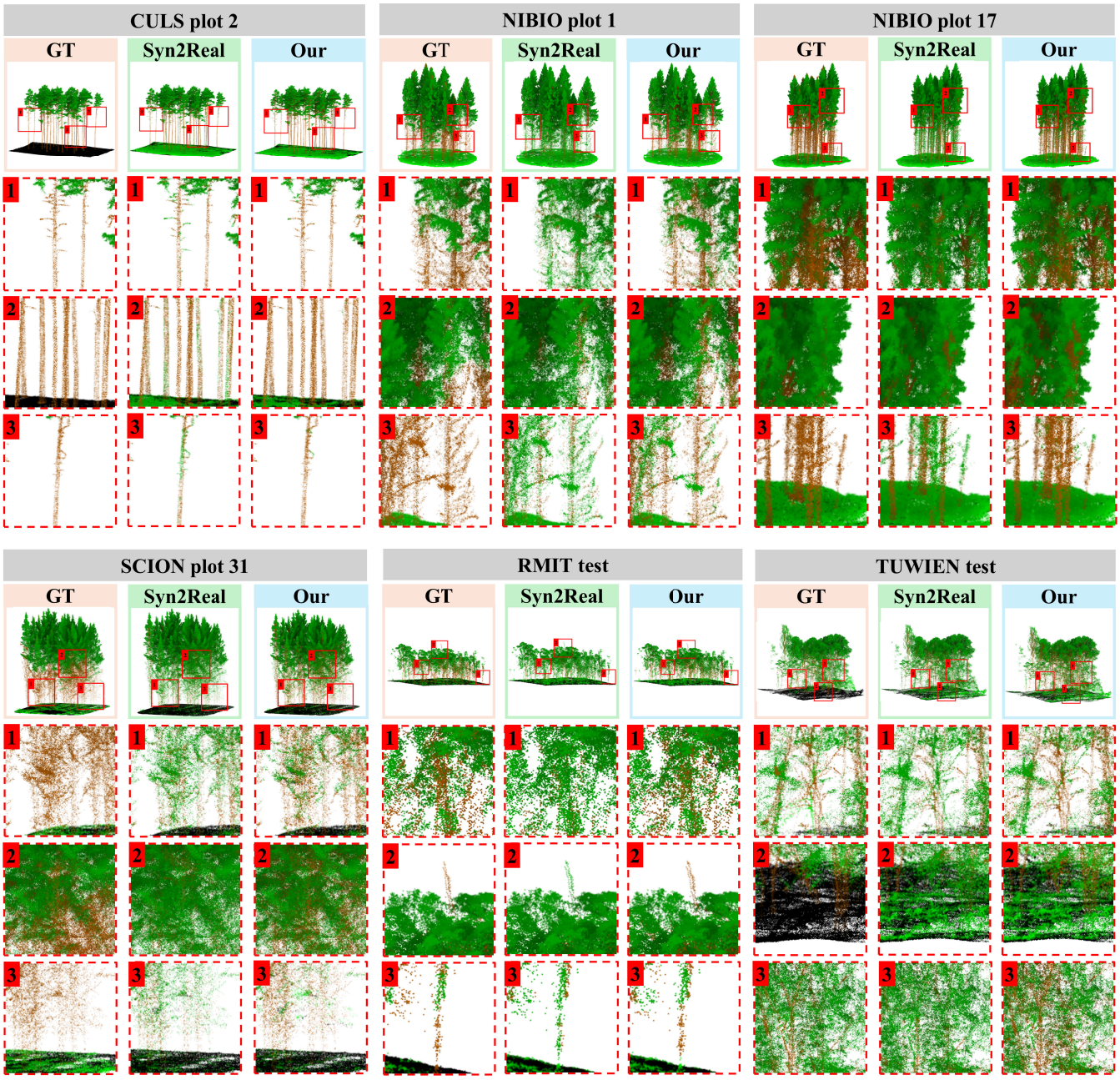


Fig. 6. Qualitative visualization of semantic segmentation results on real forest point clouds. GT, Syn2Real, and the proposed method are shown for six representative forest plots. For each plot, the first row presents the global view, followed by three rows highlighting zoomed-in local regions for detailed comparison.

behavior is particularly evident in high-density canopy regions, where stem structures are partially occluded by surrounding leaves. The proposed method, however, is able to identify plausible woody structures embedded within dense leaf points. For example, in the third local region of the NIBIO plots 1 and 17, our method separates stem points from surrounding foliage, yielding a clearer delineation between woody interiors and leaf-dominated exteriors. A similar pattern is observed in the first local region of the SCION plot, where woody components are distinctly recovered from dense canopy regions.

These qualitative results demonstrate that the proposed method yields more structurally consistent segmentation of woody components compared to direct Syn2Real transfer. The observed visual improvements are consistent with the quantitative results reported in Section IV, where the woody class exhibits the largest performance gain among all semantic categories.

D. Additional Validation on SegmentedForests

To further examine the transferability of ForestPlotUDA under different acquisition conditions, we conduct additional experiments on the SegmentedForests dataset. Different from FOR-instance, SegmentedForests consists of high-density

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TABLE III

SEMANTIC SEGMENTATION RESULTS ON THE SEGMENTEDFORESTS DATASET. THE COMPARISON EVALUATES TRANSFERABILITY FROM SYNTHETIC MLS SOURCE DATA TO HIGH-DENSITY GROUND-BASED TLS/MLS TARGET DATA

Setting	mIoU	mAcc	OA	Terrain	Understory	Wood	Leaf
Syn2Real	55.18	74.42	74.04	85.63	26.94	55.07	53.06
Ours	56.48	75.56	75.08	87.23	28.65	55.30	54.72

TABLE IV

ABLATION STUDY ON DIFFERENT MODULES OF FORESTPLOTUDA

Setting	OA	mAcc	mIoU
Syn2Real (baseline)	83.84	65.72	54.71
Weight	85.72	66.66	56.90
CGPS-only	**86.33** (+2.49)	72.50 (+6.78)	61.27 (+6.56)
CGPS + PDNorm	86.25 (+2.41)	73.03 (+7.31)	61.48 (+6.77)
Our (CGPS + PDNorm + CAMB)	85.58 (+1.74)	**73.82** (+8.10)	**61.50** (+6.79)

ground-based TLS/MLS forest point clouds, where stems, understorey vegetation, and near-ground structures are more densely captured. This setting introduces a different target-domain distribution from UAV-LS data, involving changes in scanning geometry, point density, occlusion patterns, and semantic composition.

Table III reports the comparison between direct Syn2Real transfer and the proposed method. The *Syn2Real* baseline directly transfers the model pretrained on Boreal3D-MLS to SegmentedForests, while *Ours* performs unsupervised adaptation using unlabeled SegmentedForests point clouds. The results show that the proposed method consistently improves the overall segmentation performance compared with direct transfer, increasing mIoU from 55.18% to 56.48%, mAcc from 74.42% to 75.56%, and OA from 74.04% to 75.08%. Class-wise results show clearer gains on terrain, understory, and leaf, while the wood class shows only a marginal improvement. These results suggest that the proposed strategy retains transferability to high-density TLS/MLS forest scenes, although its adaptation capability remains limited in this setting.

E. Ablation Study and Sensitivity Analysis

1) Ablation Study: An ablation study was conducted to examine the contribution of each component in the proposed framework, with quantitative results reported in Table IV. The Syn2Real setting serves as the baseline, where the model is adapted from synthetic data to real forest point clouds without explicit mechanisms for addressing class imbalance or plot-level distributional heterogeneity. Under this baseline, the model achieves an OA of 83.84%, an mAcc of 65.72%, and an mIoU of 54.71%. The weighted UDA variant (weight) is reported as an auxiliary reference, demonstrating that simple class-aware reweighting already leads to consistent improvements over Syn2Real, particularly in mAcc and mIoU.

All subsequent ablation settings are directly compared against the Syn2Real baseline. The CGPS-only configuration yields clear improvements across all evaluation metrics, with OA, mAcc, and mIoU increasing by 2.49%, 6.78%, and 6.56%, respectively. This result indicates that enhancing the quality and selection of pseudo-labels is a key factor for improving adaptation performance in forest point clouds. When PDNorm is further introduced (CGPS + PDNorm), additional gains are observed in mAcc and mIoU, while OA remains comparable to the CGPS-only setting. This behavior suggests that PDNorm primarily facilitates more balanced class-wise learning across heterogeneous forest plots, rather than optimizing overall point-wise accuracy dominated by majority classes. The full model, integrating CGPS, PDNorm, and CAMB, achieves the best performance in terms of mAcc (73.82%) and mIoU (61.50%), corresponding to improvements of 8.10 and 6.79 points over the Syn2Real baseline. Although the OA is slightly lower than that of CGPS-only, it remains consistently higher than the baseline. This trend reflects the intrinsic imbalance of forest point clouds, where improvements in minority semantic components may lead to minor tradeoffs in OA.

TABLE V

EFFECT OF THE NUMBER OF SELECTED PSEUDO-LABELED POINTS PER CLASS DURING UDA

Setting	OA	mAcc	mIoU	Terrain	Understory	Wood	Leaf
1point	85.04	71.89	59.62	44.28	67.76	37.79	88.65
5point	85.58	73.82	61.50	44.26	73.07	40.90	87.78
10point	86.10	71.91	60.97	45.40	72.56	37.07	88.86
1000point	84.78	73.46	60.66	45.79	72.61	37.60	86.62

2) Influence of Selected Pseudo-Label Quantity: Table V reports the segmentation performance under different pseudo-label selection settings, where a fixed number of high-confidence points per class are retained during UDA. The results show that the proposed framework achieves competitive performance even when extremely sparse pseudo-label supervision is provided. With only one pseudo-labeled point per class, the model already reaches an mIoU of 59.62%, indicating that a very limited number of reliable supervisory signals is sufficient to guide the adaptation process. Increasing the number of selected points to five per class further improves performance, yielding the highest mIoU of 61.50% and the highest mean class accuracy. This setting consistently provides a favorable balance between supervision strength and pseudo-label reliability. In contrast, further increasing the number of selected pseudo-labeled points does not lead to additional performance gains. When 10 or 1000 points per class are used, mIoU slightly decreases to 60.97% and 60.66%, respectively. This degradation suggests that incorporating a larger number of pseudo-labeled points introduces additional noise, which can offset the benefits of denser supervision. At the class level, a similar pattern is observed. Terrain, understory, and woody components achieve their best performance with a small number of selected points, while performance saturates or declines as more points are included. Notably, woody component segmentation is particularly sensitive to pseudo-label quality, exhibiting a clear performance drop when the number of selected points exceeds five per class. In contrast, leaf segmentation remains relatively stable across all settings, reflecting both its dominance in the data distribution and its comparatively lower sensitivity to pseudo-label noise.

3) Training Stability Analysis With the Memory Bank: To account for the randomness introduced by model initialization,

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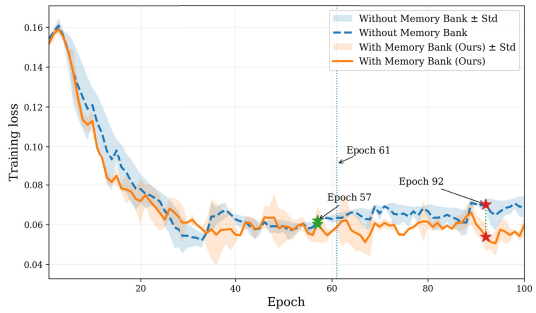


Fig. 7. Training stability analysis of the proposed framework with and without the memory bank. The solid curves denote the mean loss over three random seeds, and the shaded regions denote the standard deviation.

TABLE VI

TRAINING STABILITY ANALYSIS BASED ON ROLLING LOSS VARIANCE. LOWER VALUES INDICATE MORE STABLE OPTIMIZATION DYNAMICS

Method	Mean Loss Variance ↓	Std ↓
Without Memory Bank	3.86×10^{-4}	3.15×10^{-4}
With Memory Bank (Ours)	$\mathbf{3.33 \times 10^{-4}}$	$\mathbf{2.56 \times 10^{-4}}$

data sampling, and pseudo-label generation, all experiments were repeated three times using different random seeds, and the reported training loss curves were obtained by averaging the corresponding runs. Fig. 7 compares the training loss trajectories with and without the memory bank. During the early stage of training, approximately before 20 epochs, both variants exhibit a rapid decrease in training loss, indicating comparable initial convergence behavior. In the mid-training stage, roughly between 20 and 60 epochs, the two curves remain close to each other and intersect at several points, with minor fluctuations observed in both settings. Around Epoch 57, the advantage of the memory bank begins to emerge, as the loss gap between the two variants becomes consistently positive. Epoch 61 marks the beginning of the late-stage comparison interval, where the variant with the memory bank maintains a lower and more stable loss trajectory. The largest separation is observed around Epoch 92, indicating the most pronounced late-stage advantage of the memory bank.

Beyond visual inspection of the loss curves, we quantitatively evaluate training stability by computing the rolling loss variance with a sliding window of 10 training steps. As summarized in Table VI, incorporating the memory bank reduces the mean rolling loss variance from 3.86×10^{-4} to 3.33×10^{-4}, and decreases the standard deviation from 3.15×10^{-4} to 2.56×10^{-4}. These results suggest that aggregating historical feature information through the memory bank helps mitigate abrupt optimization fluctuations caused by noisy pseudo-labels during self-training. Consequently, the proposed framework demonstrates smoother optimization dynamics and improved training stability.

4) Effect of Confidence Thresholds: To investigate the effect of the confidence threshold on model performance, we conduct an ablation study by varying the threshold from 0.5 to 0.9. The evaluation is performed using overall metrics, including mIoU, mAcc, and OA, as well as per-class IoU for each semantic category.

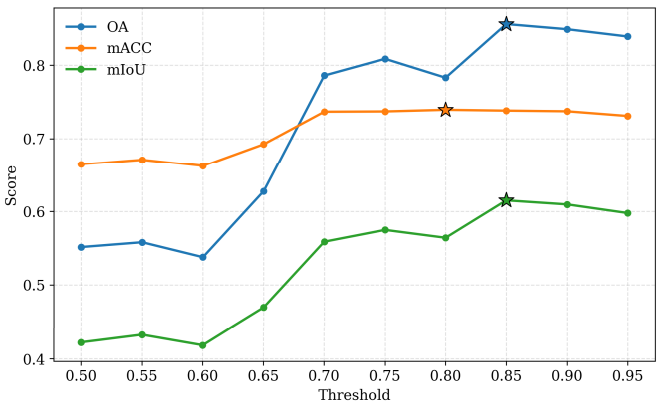


Fig. 8. Overall performance under different confidence thresholds, including mIoU, mAcc, and OA. The maximum value of each metric is highlighted with a star.

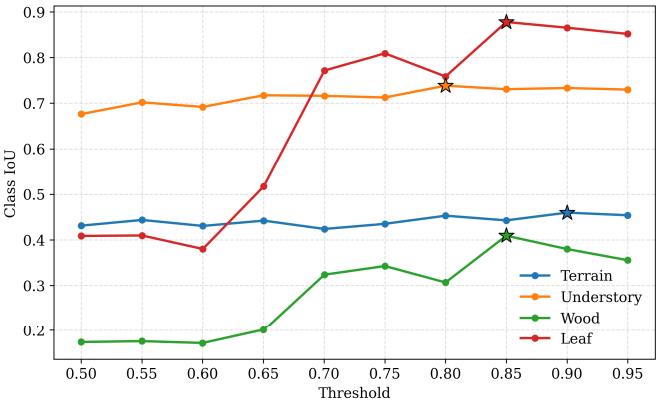


Fig. 9. Per-class IoU under different confidence thresholds for terrain, understory, wood, and leaf. The maximum IoU of each class is highlighted with a star.

As shown in Fig. 8, the overall performance exhibits a clear dependency on the confidence threshold. When the threshold is low, the model performance is significantly degraded, indicating that a large number of unreliable predictions are incorporated during training. As the threshold increases, both mIoU and OA improve steadily and reach their maximum at a threshold of 0.85, suggesting an effective balance between prediction reliability and supervision coverage. In contrast, mAcc achieves its highest value at a slightly lower threshold of 0.80 and remains relatively stable for higher thresholds. When the threshold exceeds 0.85, a slight performance degradation can be observed across OA, mAcc, and mIoU, implying that overly strict thresholds may limit the amount of effective supervision.

Fig. 9 illustrates the per-class IoU variations with respect to different confidence thresholds. The terrain and wood classes show a consistent improvement as the threshold increases and achieve their best performance at 0.85. These classes are generally more challenging due to complex geometric structures and class imbalance, and thus benefit more from higher-confidence predictions. The understory class follows a similar trend, with its highest IoU observed at a threshold of 0.80. For the leaf class, the highest IoU is obtained at a threshold

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of 0.90, while the performance at 0.85 remains comparable. Although the optimal threshold varies slightly across classes, a threshold of 0.85 provides consistently strong performance across all categories and yields a favorable tradeoff between robustness and coverage.

V. Discussion

A. Impact of Plot-Dependent Class Distributions on Unsupervised Adaptation

The differences in point-wise class distributions across forest plots reflect intrinsic variations in forest structure. This natural imbalance results in some classes, such as wood and understory, being underrepresented or absent in certain plots, as illustrated in Fig. 4. This plot-dependent class imbalance affects UDA. When pseudo-labeled samples from multiple plots are aggregated for training (mixed plot), the combined class distributions include both dominant and rare classes, which can produce inconsistent pseudo-labels and variable gradient signals. Under these conditions, training does not reach a stable converged state. In contrast, when adaptation is conducted independently for each plot (*single plot*), the class distributions within each batch are more consistent, supporting reliable pseudo-label generation and stable convergence, as reflected in the reported plot-level mIoU values in Table I.

These observations have implications for algorithm design in forest point cloud segmentation. They highlight the importance of addressing plot-specific distributional heterogeneity, either through separate adaptation for each plot or through methods that explicitly handle class imbalance. Approaches such as class-aware pseudo-label selection, per-plot normalization, or plot-conditioned adaptation modules may improve the robustness of self-training frameworks. Furthermore, evaluation of forest point cloud algorithms should take into account both intraplot imbalance and interplot heterogeneity, as these factors influence convergence behavior and segmentation performance.

B. Limitations of Existing Unsupervised Adaptation Strategies in Forest Scenes

The comparative evaluation in Section IV-B further indicates that existing UDA and data mixing strategies, such as CosMix and PolarMix, struggle to generalize to plot-level forest point cloud segmentation. While these methods have shown effectiveness in urban or object-centric scenarios, their assumptions regarding spatial continuity, class balance, and structural regularity are often violated in forest environments. The resulting instability and negative transfer observed in our experiments suggest that such approaches may be inherently sensitive to the extreme class imbalance and structural complexity of forest scenes.

Even when incorporating class-aware weighting during self-training, performance improvements remain limited, particularly for woody components. This suggests that simple reweighting schemes are insufficient to compensate for the compounded effects of domain shift and class sparsity. In contrast to foliage, which dominates the point distribution and exhibits relatively stable performance across methods, minority classes such as terrain and woody components require more robust adaptation mechanisms that go beyond point-level reweighting or naive data mixing.

C. Transferability to High-Density Ground-Based Forest Point Clouds

The additional experiments on SegmentedForests provide a complementary evaluation of ForestPlotUDA on high-density ground-based forest point clouds. In this experiment, Boreal3D-MLS is used as the synthetic source domain, and SegmentedForests is used as the real target domain. This setting evaluates Syn2Real adaptation from MLS-style simulated forest point clouds to real TLS/MLS forest point clouds. ForestPlotUDA improves direct Boreal3D-MLS to SegmentedForests transfer from 55.18% to 56.48% mIoU, from 74.42% to 75.56% mAcc, and from 74.04% to 75.08% OA. These results show that the proposed framework retains transferability to high-density ground-based forest point clouds. However, the improvement is relatively modest, indicating that its adaptation capability in this setting remains limited.

This limited improvement may be related to three factors. First, the dominant error patterns in high-density TLS/MLS point clouds can differ from those in UAV-LS data. Dense local structures, near-ground clutter, stem-branch interactions, and fine-grained geometric variations may require stronger exploitation of local geometry, while the current sparse point selection strategy may not fully capture these dense structural details. Second, the semantic labels of SegmentedForests are not fully aligned with those of the synthetic Boreal3D-MLS source domain. Although both datasets are mapped into the same four-class setting, the original category definitions and label boundaries differ, especially for terrain, understory, wood, and leaf. Such source-target label discrepancies can affect pseudo-label reliability and limit the effectiveness of self-training. Third, a Syn2Real gap still exists between Boreal3D-MLS and real TLS/MLS forest point clouds, including differences in scanning trajectories, occlusion patterns, point density distributions, sensor noise, and annotation conventions.

D. Effectiveness of Sparse Pseudo-Label Selection

The ablation results in Section IV-E demonstrate that effective UDA can be achieved with a remarkably small number of carefully selected pseudo-labeled points per class. The observation that segmentation performance peaks when only a few high-confidence points are used suggests that the quality of pseudo-labels is more critical than their quantity. Increasing the number of pseudo-labeled points introduces a higher likelihood of noisy labels, which can obscure useful supervisory signals and degrade model performance, particularly for structurally ambiguous classes such as woody components.

This behavior aligns with the inherent uncertainty of pseudo-label generation in complex forest environments. In the presence of occlusion, overlapping vegetation layers, and uneven point densities, even moderately confident predictions may still be unreliable. Consequently, aggressively expanding pseudo-label sets can amplify errors rather than reduce them.

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The relative stability of foliage segmentation across different pseudo-label settings further reflects its dominance and structural consistency, whereas minority classes remain more sensitive to label noise.

E. Implications for Plot-Level Forest Point Cloud Segmentation

Our results yield practical recommendations for plot-level semantic segmentation of forest point clouds. Rather than aggregating pseudo-labeled samples across heterogeneous plots, effective adaptation benefits from conditioning the learning process on local plot characteristics. In particular, plot-conditioned feature normalization and adaptation help mitigate distributional mismatches arising from varying point densities, structural complexity, and occlusion patterns across plots. In addition, the results highlight the importance of prioritizing pseudo-label quality over quantity. Using a sparse set of high-confidence pseudo-labeled points produced more reliable supervision in our experiments (optimal setting: 5 points per class at confidence $\geq$0.85; see Table V), reducing noisy gradient updates compared with denser pseudo-labeling.

Under these design choices, the proposed Syn2Real adaptation framework demonstrates an improved ability to recover woody components that are frequently misclassified as foliage in baseline transfer settings, especially in dense canopy regions where structural cues are easily obscured. These improvements indicate that combining plot-level conditioning with selective pseudo-labeling is an effective strategy for addressing minority-class degradation in forest scenes. Nevertheless, segmentation performance for extremely sparse or heavily occluded classes remains limited, suggesting that additional structural priors or more advanced adaptation mechanisms will be necessary to further improve robustness across diverse forest conditions.

F. Limitations and Future Work

Despite the promising results, several limitations remain. First, although the main experiments focus on UAV-LS forest point clouds, the additional SegmentedForests validation provides preliminary evidence of transferability to ground-based TLS/MLS data. However, this cross-platform evaluation remains limited to one additional benchmark and a harmonized four-class label setting, and more comprehensive evaluation across different sensing platforms, forest types, and annotation protocols is still needed. Second, while synthetic data provide accurate and scalable annotations, a domain gap remains between simulated and real forest point clouds, particularly under complex occlusion patterns and acquisition-specific point density variations. Third, the current framework relies on plot-level or acquisition-level identity information for plot-conditioned normalization, and its behavior under incomplete or unavailable plot metadata should be further investigated. Future work will explore more flexible adaptation strategies, label-aware cross-dataset harmonization, and stronger structural priors to further improve robustness across diverse forest conditions.

VI. Conclusion

This study addresses the challenging Syn2Real domain transfer for plot-level forest point clouds semantic segmentation, where structural heterogeneity and class imbalance severely hinder direct model generalization. To bridge this gap, we propose a specific self-training framework that aligns feature distributions using sparse, class-balanced pseudo-labels. On the FOR-instance benchmark, our method achieves 61.50% mIoU, surpassing the direct transfer baseline by 6.79 points and yielding a substantial 18.08-point improvement on the difficult woody class. Additional validation on SegmentedForests further shows consistent improvements over direct transfer under high-density ground-based TLS/MLS forest point clouds. These results confirm that selective, class-aware adaptation effectively unlocks the potential of synthetic data, offering a scalable, annotation-free solution for real-world forest analysis.

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